

Ex.10 Implement solution for dining Philosopher problem

Implement solution for dining Philosopher problem.

AIM:

To implement solution for dining Philosopher problem.

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <semaphore.h>

#define NUM_PHILOSOPHERS 5

sem_t forks[NUM_PHILOSOPHERS];
sem_t mutex;

void take_fork(int id) {
    sem_wait(&mutex);
    sem_wait(&forks[(id + 1) % NUM_PHILOSOPHERS]);
    sem_wait(&forks[id]);
    sem_post(&mutex);
}

void put_fork(int id) {
    sem_post(&forks[id]);
    sem_post(&forks[(id + 1) % NUM_PHILOSOPHERS]);
}

void test_blocked_philosophers() {
    for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
        if (sem_trywait(&forks[i]) == 0) {
            printf("Philosopher %d is not blocked.\n", i);
            sem_post(&forks[i]);
        } else {
            printf("Philosopher %d is blocked.\n", i);
        }
    }
}
```

```

    }
    printf("\n\n");
}

int main() {
    int ch;
    sem_init(&mutex, 0, 1);

    for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
        sem_init(&forks[i], 0, 1);
    }

    printf("Initial state:\n");
    test_blocked_philosophers();
    printf("\n");

    printf("With deadlock in the execution (1/0): ");
    scanf("%d",&ch);

    for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
        if(ch==0) {
            printf("Philosopher %d wants to take forks:\n", i);
            take_fork(i);
            sleep(1);
            printf("Philosopher %d took forks.\n\n", i);
            printf("... State after taking forks ...\n");
            test_blocked_philosophers();
            sleep(1);
            printf("\n");

            printf("Philosopher %d releases forks:\n", i);
            put_fork(i);
            sleep(1);
            printf("Philosopher %d released forks.\n\n", i);
            printf("... State after releasing forks ...\n");
            test_blocked_philosophers();
            sleep(1);
            printf("\n");
        }
        else
        {
            printf("Philosopher %d wants to take forks:\n", i);
            take_fork(i);
            sleep(1);

```

```

printf("Philosopher %d took forks.\n\n", i);
printf("... State after taking forks ...\n");
test_blocked_philosophers();
sleep(1);
printf("\n");

int next_phil = (i + 1) % NUM_PHILOSOPHERS;
printf("Philosopher %d wants to take forks:\n", next_phil);
take_fork(next_phil);
sleep(1);
printf("Philosopher %d took forks.\n\n", next_phil);
printf("... State after taking forks ...\n");
test_blocked_philosophers();
sleep(1);
printf("\n");
}
}

sem_destroy(&mutex);
for (int i = 0; i < NUM_PHILOSOPHERS; i++) {
    sem_destroy(&forks[i]);
}
return 0;
}

```

SCREENSHOT: Without deadlock:

```

Thu 09:53
guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
collect2: error: ld returned 1 exit status
guest-9cp7kb@admin25-VirtualBox:~$ gcc -o DP DP.c -pthread
guest-9cp7kb@admin25-VirtualBox:~$ ./DP
Initial state:
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

With deadlock in the execution (1/0): 0
Philosopher 0 wants to take forks:
Philosopher 0 took forks.

... State after taking forks ...
Philosopher 0 is blocked.
Philosopher 1 is blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 0 releases forks:
Philosopher 0 released forks.

... State after releasing forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 1 wants to take forks:
Philosopher 1 took forks.

```

```
Activities Terminal Thu 09:53
guest-9cp7kb@admin25-VirtualBox: ~

File Edit View Search Terminal Help
Philosopher 1 wants to take forks:
Philosopher 1 took forks.

... State after taking forks ...
Philosopher 0 is not blocked.
Philosopher 1 is blocked.
Philosopher 2 is blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 1 releases forks:
Philosopher 1 released forks.

... State after releasing forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 2 wants to take forks:
Philosopher 2 took forks.

... State after taking forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is blocked.
Philosopher 3 is blocked.
Philosopher 4 is not blocked.

Philosopher 2 releases forks:
Philosopher 2 released forks.
```

```
Activities Terminal Thu 09:54
guest-9cp7kb@admin25-VirtualBox: ~

File Edit View Search Terminal Help
Philosopher 2 releases forks:
Philosopher 2 released forks.

... State after releasing forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 3 wants to take forks:
Philosopher 3 took forks.

... State after taking forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is blocked.
Philosopher 4 is blocked.

Philosopher 3 releases forks:
Philosopher 3 released forks.

... State after releasing forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 4 wants to take forks:
Philosopher 4 took forks.
```

```
Activities Terminal Thu 09:55
guest-9cp7kb@admin25-VirtualBox: ~

Philosopher 3 releases forks:
Philosopher 3 released forks.

... State after releasing forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 4 wants to take forks:
Philosopher 4 took forks.

... State after taking forks ...
Philosopher 0 is blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is blocked.

Philosopher 4 releases forks:
Philosopher 4 released forks.

... State after releasing forks ...
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

guest-9cp7kb@admin25-VirtualBox:~$
```

With deadlock:

```
Activities Terminal Thu 10:00
guest-9cp7kb@admin25-VirtualBox: ~

guest-9cp7kb@admin25-VirtualBox:~$ ./DP
Initial state:
Philosopher 0 is not blocked.
Philosopher 1 is not blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

With deadlock in the execution (1/0): 1
Philosopher 0 wants to take forks:
Philosopher 0 took forks.

... State after taking forks ...
Philosopher 0 is blocked.
Philosopher 1 is blocked.
Philosopher 2 is not blocked.
Philosopher 3 is not blocked.
Philosopher 4 is not blocked.

Philosopher 1 wants to take forks:

```

RESULT:

Thus implemented the dining philosopher problem.